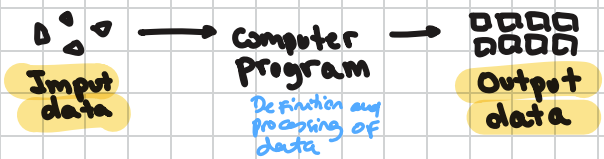
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BIOINFORMÁTICA : R

R is a interpreted language → obtain descriptive Statistics.
(media, correlation...)
hypothesis testing
Stat inference
Data virualization
Predictive analysis

Basic Data Types :

All programs are designed to



Data has many types → numbers, Text, Dates, Factors (...)

They need to be Specifically defined and processed by programming languages to achieve the correct result.

Common data types in R

1. Integer (whole numbers) → Written with an "L" Sufix (L) -1L, 0L, 1L...
(-1, -2, 0, 1, 2...)

class(1L) → 'integer'
is.integer(1L) → TRUE

Te dices que el numero es un integer

Para preguntar si el (no) es integer

2. Numeric (Integer + decimals)
(-1.5, -1.4, -1, 0, 1, 1.2...)

class(1) → 'numeric'
is.numeric(1) → TRUE

3. Textual data (characters)
("A", "I like R", "kst@kst.com")

class("A") → "character"
is.character("kst@kst.com") → TRUE

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4. Boolean data "logical" (TRUE, FALSE)
   - class(TRUE) → 'logical'
   - is.logical(TRUE) → TRUE

5. Complex: 1+2i

6. Raw (binary): 1c 2b 4f

CONVERSION of DATA

1. Integer to numeric: as.numeric(2L) → 2 (numeric)
   (2L: integer)

2. Numeric to integer: as.integer(2.2) → 2 (integer)
   (2.2: numeric)

3. Numeric to character: as.character(3) → '3' (character)
   (3: numeric)

4. Character to numeric: as.numeric("3") → 3 (numeric)
   ("3": character)
   - as.numeric("A") → ERROR (NA induced by coercion)

5. Numeric to logical: as.logical(1) → TRUE (logical)
   (1: any number ≠ 0)
   as.logical(0) → FALSE (logical)
   (0: only w/ 0)

6. Character to logical: as.logical("FALSE") → FALSE (logical)

7. Logical to numeric: as.numeric(FALSE) → 0 (numeric)

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BASIC MATH

Addition: 139 + 121
Division: 260 / 60

Substraction: 1 - 1
Multiplication: 4 * 2
Exponentiation: 4 ^ 2

If we want to SAVE the results, we need to store them in a variable:

X <- 139 + 121

That way, when we type X → 260

WE CAN ALSO USE = instead of <-

If we want to REVISION a variable simply do it all over again.

Old variable : x = 260 (old)

Rewrite it : x <- x/60

x = 4.333 (New)

To remove variables from memory use:

rm(my_variable)

VARIABLE NAMES { total time → total
We can use whatever we want but sometimes its easier to remember if we follow this: { total hours → total_hr

Order of operations → Remember parentheses (1+1) / 2

1° 2°

STRINGS

String: Secuence of 1 or more characters Ex: "Toy story"

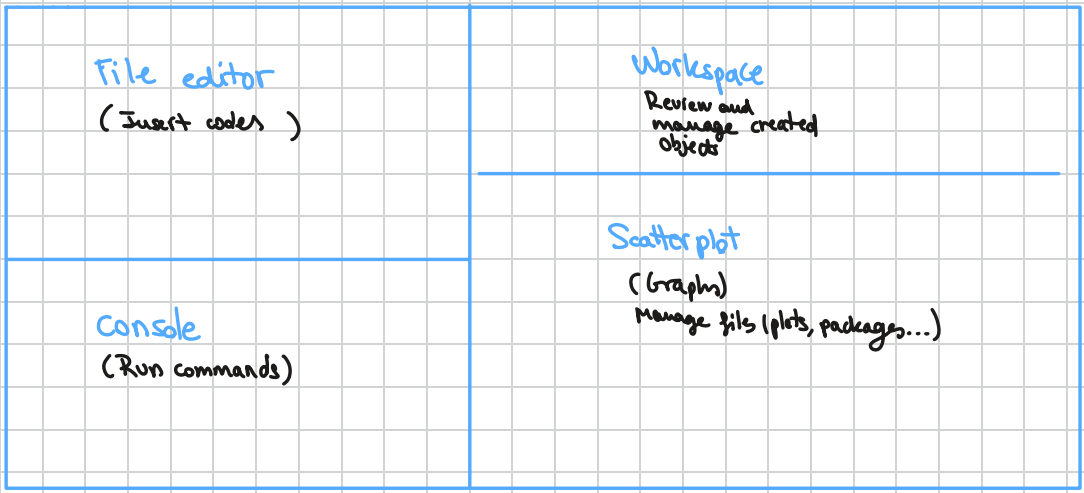
To create a string you need to enclose the characters in: " "

movie <- "Toy story"

variable — necessary to create string

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R environment



To delete everything: Ctrl + L

print ("....")

X = "Hello world"
y = "Hello back"
print (x) + print (y) → "Hello world"
"Hello back"

Create R script file → type getwd() in console

Run code → Select ALL lines and then RUN icon
↳ OR RUN THE CURRENT LINE OR SELECTION

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